

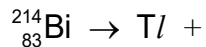
9 (a) Explain what is meant by the *half-life* of a nuclide.

.....
 [1]

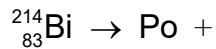
(b) Bismuth-214 ($^{214}_{83}\text{Bi}$) has an unusual decay pattern. Sometimes it decays by emission of an α -particle into thallium (Tl) and sometimes by emission of a β^- -particle into polonium (Po).

(i) Complete the nuclear equations for these two decays of $^{214}_{83}\text{Bi}$.

α -decay:



β^- -decay:



[2]

(ii) Data for the α -decay of $^{214}_{83}\text{Bi}$ to form Tl are given in Fig. 9.1.

nucleus	mass of nucleus / u
Bi	213.9987
Tl	209.9901
α	4.0015

Fig. 9.1

Use the data in Fig. 9.1 to calculate the energy released during the decay.

energy released = J [3]

(iii) The two decay patterns of $^{214}_{83}\text{Bi}$ each give rise to γ -ray photons.

Suggest why photons from these two decays have different energies.

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 [2]

(c) (i) On Fig. 9.2, sketch a labelled graph (without numerical values) to show how starting with only $^{214}_{83}\text{Bi}$ nuclei, the activity of the $^{214}_{83}\text{Bi}$ changes with time. Label the graph Bi. [1]

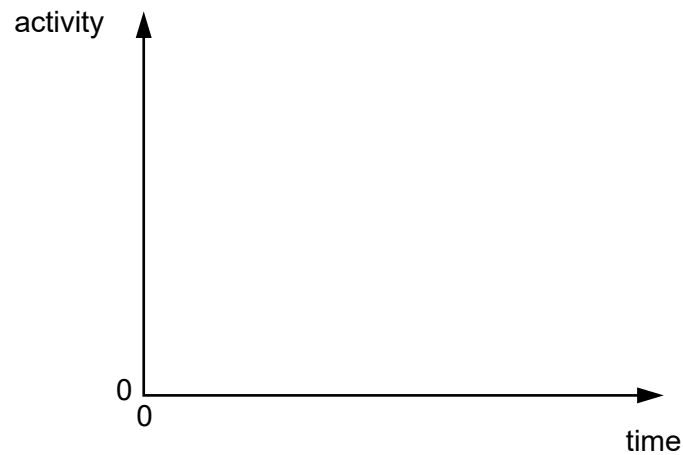


Fig. 9.2

(ii) The thallium (Tl) produced is also radioactive and has a shorter half-life than that of $^{214}_{83}\text{Bi}$.

On the same axes in Fig. 9.2, sketch another graph to show how the activity of the Tl changes with time. Label the graph Tl. [1]

(d) A sample of $^{214}_{83}\text{Bi}$ has mass $2.0\text{ }\mu\text{g}$ at time $t = 0\text{ s}$.

(i) Show that the sample contains approximately 5.6×10^{15} atoms.

[1]

(ii) The activity at $t = 0\text{ s}$ is $3.3 \times 10^{12}\text{ Bq}$.

Calculate the decay constant.

decay constant = s^{-1} [2]

(iii) Calculate the half-life.

half-life = s [1]

(iv) Calculate the time taken for the activity to fall to $3.3 \times 10^6\text{ Bq}$.

time = s [2]

- (e) The $^{214}_{83}\text{Bi}$ sample is stored in a non-radioactive lead container with thin walls of thickness 1 mm. Background radiation is negligible.

Suggest why, although the lead container provides adequate shielding for the α and β^- -emissions, some X-ray radiation may be detected outside the lead container.

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..... [4]

End of paper